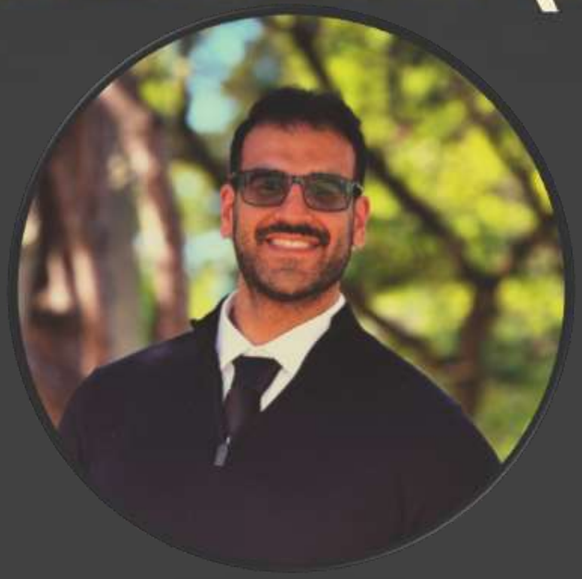
Lab2Market
Validate

Mid Program Report

AddManuChain (Alma-Tech)

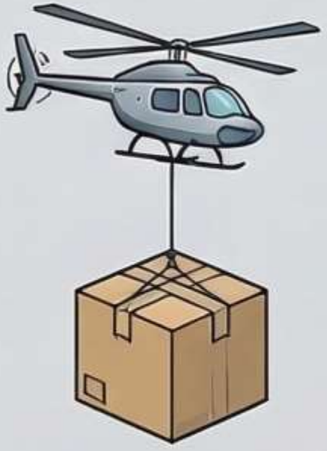


DALHOUSIE UNIVERSITY

AddManuChain: The End of the Physical Warehouse

TODAY

The High Cost of "Just in Case"



LONG DELAYS:
Missing components cause up to **$7M/day** in downtime.

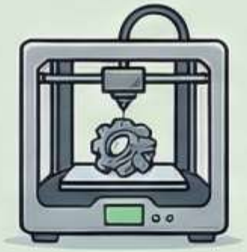
High cost of "just in case" inventory

The Digital Birth Certificate:
Every part includes a live record of design, material, and certification to guarantee safety.

WITH ADDMANUCHAIN

CERTIFIED DIGITAL LIBRARY

HOURS, NOT MONTHS



PRINT ON DEMAND, LOCALLY: Orders are routed to the nearest qualified 3D printing facility for delivery in hours.

DIGITAL LIBRARY VS. PHYSICAL WAREHOUSE:
Replace massive storage costs and emergency hellcopters with a pre-certified digital parts library.

OBSOLETE PARTS → OFFSHORE O&G → NAVAL DEFENCE → AEROSPACE

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Validate

Status	Count	% of Total
Connection	124	29.5%
2nd Request	119	28.3%
1st Request	60	14.3%
Interview Completed	**55**	**13.1%**
Dead Lead	35	8.3%
Scheduled	10	2.4%
Company Email	6	1.4%
3rd Request	1	0.2%
Total Outreach	**420**	**100.0%**

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Corporate Engagement Registry: Completed Interviews

41 Organizations Interviewed

The registry documents completed interviews across a wide spectrum of industries, ranging from global energy leaders to niche manufacturing innovators.

Energy & Resources

- ExxonMobil Canada
- AkerSolutions
- equinor Canada Ltd
- Frobisher Energy Services
- Suncor
- OMV
- Baker Hughes
- Agnico Eagle
- Noble Corporation

Maritime & Logistics

- msc
- CSL Group Inc
- Horizon Maritime Services
- Irving Shipbuilding / Cove
- Clearwater Seafoods
- Atlantic XL
- Maritime Launch Services

Manufacturing & Tech

- Equispheres
- CAMufacturing Solutions Inc
- Galaxia
- Reaction Dynamics
- Tronosjet
- Agile Manufacturing Inc
- Proax
- Maritime Launch Services
- Design Smith Inc.
- Next Generation Manufacturing Canada (NGen)
- MAN Energy Solutions Canada
- Thales
- PolyUnity Tech Inc.
- Enginuity

Engineering & Consulting

- Lloyd's Register
- intecsea (Worley)
- Deloitte.
- The Assembly Studio
- Shield Group of Companies

Government & Public Sector

- Government of Nova Scotia
- CME Canadian Manufacturers & Exporters
- National Research Council Canada
- InnoTech Alberta

Academic & Training

- Dalhousie University
- UNB University of New Brunswick
- CQFA Centre québécois de formation aéronautique

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Original Idea

When I look back at the beginning of the program my business thesis looked like...

Provide offshore O&G, maritime, and defence

↓

With a digital platform for on-demand 3D printing

↓

To eliminate costly warehousing and emergency logistics

↓

So operators get certified parts printed locally in hours

Current Value Proposition

After 55+ interviews, I have refined my value proposition as follows:

AddManuChain provides remote operators (O&G, maritime, Arctic mines, naval) with a pre-certified digital spare parts library and AI-powered orchestration platform connecting operators, LR/DNV-qualified AM facilities, and classification societies.

Replaces physical inventory ($30-50M/vessel)	Eliminates emergency hot-shotting	Cuts downtime costs ($50K-$7M/day)

The platform's primary role is NOT AM matchmaking—it is closing the certification permission gap. Industry sees value. Technical capability exists. Regulatory permission is the missing piece. AddManuChain's Digital Birth Certificate bridges that gap.

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TARGET SEGMENTS & EARLY ADOPTERS

Offshore Oil & Gas Operators (MRO supply managers, R&D tech advisors)	Maritime Fleet Operators (Vessel superintendents, in-service support)	Naval Defence / Shipbuilding (Innovation teams, Navy procurement)
Equinor (Bay du Nord $14B greenfield) ExxonMobil Canada ($17.5M SMART Lab)	Clearwater Seafoods (using 3D printers on vessels) MAN Energy Solutions	Thales Canada / COVE (running AM challenge) Irving Shipbuilding

Severity: HIGH	**Certification & regulatory permission gap:** Classification societies lagging; 2+ **year** minimum certification timelines; operators fear liability.
Severity: HIGH	**Excessive physical inventory & emergency logistics:** $30–50M tied-up capital per vessel; $50K–$7M/day downtime.
Severity: HIGH	**Fragmented supply chain visibility:** MRO supply chains are last-minute; massive manual SAP/EWM intervention required.

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PURCHASING DECISION

Multi-Stakeholder Buying Process for Digital Inventory Adoption

Stage 1: Problem Recognition

Who: Ops/Maintenance Engineers

Trigger: Part failure on remote asset.

Insight: *Pain is normalized; triggered by failure.*

Stage 2: Internal Champion

Who: R&D / Digitalization Managers

Gate: Must demonstrate ≥10–15% ROI.

Stage 3: Certification

Who: Classification Societies (LR, DNV, ABS)

Gate: No adoption without classification sign-off.

Insight: *The 'Permission' Gap.*

Stage 4: Procurement / Budget

Who: VP Operations, CFO

Gate: Annual budget cycle alignment.

Stage 5: Operational Deployment

Who: Facility operators, vessel crew

Insight: *Governance must be neutral.*

FOUNDATIONAL STAKEHOLDER: THE OEM

They control IP. Can block sharing or willing license to outsource non-core service. Obsolete parts = lowest friction entry point.

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Validated Hypotheses	Invalidated Hypotheses	Hypotheses To Be Tested
✓ **Pain is real/severe:** $30–50M inventory/vessel, $50K–$7M/day downtime.	✗ **AM directly enables digital inventory:** False. It is a distinct concept.	? **Will OEMs license designs via 3rd party?** Unconfirmed directly with OEMs.
✓ **Certification is #1 barrier:** 2+ year timelines; no universal standards.	✗ **Irving manages AOPS spare parts:** False. Thales won the contract.	? **In-process monitoring vs. post-test?** Could slash per-order cert costs.
✓ **Digital inventory is feasible:** SAP handles it; MAN Energy uses it.	✗ **AI is a strong selling point:** False. Operators want invisible, proven solutions.	? **Does 'Aerospace First' cert transfer?** Will O&G/Maritime accept it?
✓ **IP security is a top barrier:** Unwilling to share IP regardless of NDAs.	✗ **Downtime cost alone motivates action:** False. Obsolescence is the true driver.	? **Willingness-to-pay?** ROI established, but pricing model untested.
✓ **Adoption is political:** Needs major-company top-down mandate.	✗ **Offshore NL needs new sharing system:** False. Informal rig-sharing works fine.	? **Differentiation from FieldNode?** Incumbent already has O&G consortium.
✓ **Obsolete parts = lowest friction:** No OEM IP conflict, lower cert complexity.		

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REFLECTION

Challenges I've Overcome

- Breaking into senior industry circles as a PhD student.
- Overcoming academic bias by leading with validated pain points.
- Navigating NDA-restricted environments by focusing on administrative pain.

My Biggest Achievement

- Conducting 55+ interviews in 6 weeks.
- Secured access to Lloyd's Register's global AM technical authority for complete certification pathway.

Biggest 'AHA' Moment

Tyler Beatty's Value-Capability-Permission framework.

Value is recognized. Capability exists. Regulatory permission is the ONLY missing piece. Shifted positioning from "AM marketplace" to "certification bridge."

Secondary: OEMs may be willing licensors looking to monetize obsolete spare parts.

What I Still Need To Overcome

- Direct OEM validation for design licensing.
- Competitive differentiation from FieldNode.
- Pricing model validation & Mining/Nuclear exploration.

What I Still Need To Improve On

- Shortening interview times (extracting key insights in 15 mins).
- Testing the specific pricing model, not just problem existence.
- Building a rigorous financial model for platform economics.

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